A Modular, FHIR-Integrated Framework for LLM-Assisted Pre-Consultation

Wei-Hsuan Tai[1,2,3†], Chun-Yi Chen[2,3†], Guan-Ying Huang[4†], Chien-Chang Lee[4,5,6*]

[1]Department of Biomedical Engineering, National Taiwan University, Taipei, Taiwan.
[2]Industry, Science and Technology International Strategy Center, Industrial Technology Research Institute, Hsinchu, Taiwan.
[3]Next-Generation Digital Health Platform Office, Ministry of Health and Welfare, Taipei, Taiwan.
[4*]Department of Emergency Medicine, National Taiwan University Hospital, Taipei, Taiwan.
[5]Department of Emergency Medicine, College of Medicine, National Taiwan University, Taipei, Taiwan.
[6]Department of Information Management, Ministry of Health and Welfare, Taipei, Taiwan.

*Corresponding author(s). E-mail(s): cclee100@gmail.com;
Contributing authors: b12508026@ntu.edu.tw; TODO; TODO;
†These authors contributed equally to this work.

Abstract

We describe a modular framework for LLM-assisted pre-consultation with questionnaire-based question selection and structured TW Core FHIR write-back. Synthetic verification maps requirements to explicit checks; clinician confirmation was simulated. Absolute requirements R0, R1, R2, R4, R6 pass; R3, R5 fail. R0 supports deterministic question selection through saved-call auditing, static code evidence, and repeated-path stability. R3 fails because written FHIR paths differ from runtime paths for free-text EHR prefill, despite agreement for patient-entered branching answers. The discrepancy reflects fixed-rule encoding differences, not AI question selection. R5 retains known scripted rule gaps. Configuration-only replacement (R7) is Pass for alternate Gemini, matching default outcomes including shared R3/R5 failures, and Fail for Gemma

because of handoffs and missed stops. National capacity (R8) is Not verifiable without demand data; exchange-hub verification (R9) is Blocked by unavailable access. Structural/profile checks use `-tx n/a`, without terminology resolution. Requirement-defined criteria were applied post hoc, distinct from prespecified targets. Findings remain provisional pending physician review and independent audits; they establish bounded software verification, not clinical validation or deployment readiness.

Keywords: FHIR, Pre-consultation, Conversational AI, Clinical Documentation

1 Introduction

Low-acuity visits place a growing burden on unscheduled care worldwide.Pines et al. (2011) In Taiwan, despite low-cost access under National Health Insurance, its emergency departments remain chronically crowded, and about one-third of visits are non-emergent.Huang et al. (2019) This crowding prolongs waits, worsens outcomes, and drives clinician burnout.Morley et al. (2018); Ghaseminejad et al. (2025)

Large language models (LLMs) now support numerous clinical tasks, including ambient scribing, EHR documentation, and clinical decision support.Nair et al. (2026) Agentic systems extend these capabilities through planning, tool use, and multi-step action, enabling closer integration with health-system workflows and conversational diagnosis.Tu et al. (2025) However, although various benchmarks now evaluate LLMs on realistic tasks and interactive encounters, most agentic systems remain confined to simulated settings.Liu et al. (2026)

Here, we present XXX, an LLM-based agent for adults presenting in Taiwanese urgent care centers. XXX conducts structured history-taking by text or voice, drafts a clinician-ready note, and supports decision-making with a ranked differential and not-to-miss conditions, alongside suggested examinations and imaging. To limit hallucination, the LLM only extracts clinical facts linked to the patient's own words, while safety screening, question selection, and diagnostic ranking follow auditable rules. XXX connects to EHRs via SMART on FHIR and was evaluated on benchmark.

2 Methods

2.1 Scope and implemented architecture

We verified a reference implementation for structured pre-consultation using synthetic patients. The outcomes concern data interchange, interview-to-write-back completion, model replacement, and local concurrency. Diagnostic and triage accuracy, clinical note quality, and comparisons with clinicians were outside scope.

The implementation combines Vue patient and clinician interfaces, a FastAPI backend, local retrieval, and a HAPI FHIR R4 sandbox (Fig. 1) (Vue.js 2026; Ramírez 2026; HAPI FHIR Project 2026; HL7 International 2019b). A patient invitation is exchanged for a scoped session cookie; the clinician launcher binds patient and encounter identifiers. The separate SMART integration surface follows a different

Table 1 Write-back resources and assertion boundaries.

Resource	Content and boundary
QuestionnaireResponse	Ordered answers, source markers, and the pinned canonical reference.
Observation	Smoking status only for an explicit mapped patient selection; EHR text is not promoted to a new patient assertion.
Composition	Preliminary narrative, confirmed sections, and links to structured answers.
Provenance	Patient informant, AI assembly, and simulated clinician-review roles; no actual clinical approval.

authorization flow (Mandel et al. 2016; HL7 International 2024). These verification activities exercise text-channel HTTP APIs, not browser/OAuth launch or voice input. Isolated verification databases and an explicit loopback development authorization bypass separate the runs from production services.

In the default questionnaire engine evaluated here, questionnaire rules select the next enabled question and skip EHR-prefilled fields; the LLM does not choose the next question. An alternative engine in the code base that makes per-turn model calls was not evaluated. Chief-complaint keywords select the questionnaire first. Only no keyword match permits evidence-grounded model extraction as a routing fallback. Rules and model extraction also screen the chief complaint; failed semantic assessment triggers manual handoff. At routine completion, retrieval supplies evidence for model-generated report sections and decision-support suggestions (Lewis et al. 2020). These suggestions are neither validated diagnoses nor published orders.

2.2 Questionnaire and write-back contracts

The encoder maps source fields to stable `linkId` values, options to coded answers, and supported conditions to `enableWhen`. Questionnaire and QuestionnaireResponse separate form definitions from answers (HL7 International 2019a, 2022). The canonical version includes a source digest and is pinned in the interview trace when a route is selected. Write-back rejects a changed canonical version; older records without a pin use current resolution with an explicit legacy tag. Application extensions carry source and presentation semantics, so generic FHIR consumers cannot assume complete runtime equivalence. The questionnaire is encoded and published as FHIR Questionnaire, while source JSON drives execution. Tested equivalence holds for patient-entered branching answers but diverges for free-text EHR prefill, as quantified by R3.

Structured write-back is opt-in; Composition-only write-back remains the default. After simulated clinician confirmation, an authorized request creates the transaction in Table 1. Stable references and conditional creation support retry idempotency. Patient, encounter, institution, and issuer checks precede persistence. No differential-diagnosis Conditions, ClinicalImpressions, ServiceRequests, or safety Flags are created.

An absent-EHR placeholder is omitted from the response. Free text outside the option grammar is represented by the existing coded “other” option plus its nested

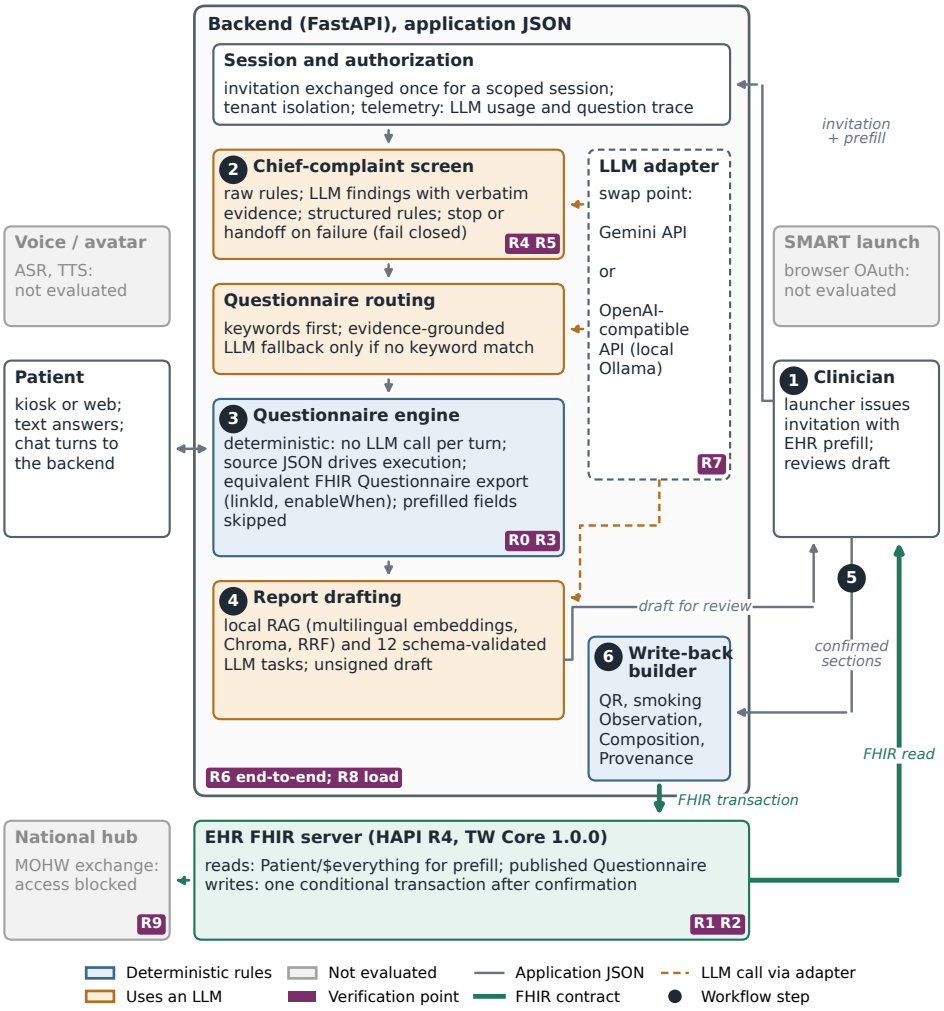


Fig. 1 Implemented modules, contracts and verification points. Workflow steps: (1) the clinician launcher reads EHR history and issues an invitation with prefill; (2) chief-complaint screening and questionnaire routing; (3) deterministic questionnaire turns; (4) report drafting; (5) clinician review and confirmation; (6) one FHIR write-back transaction. Source JSON drives the questionnaire engine and an equivalent FHIR Questionnaire is exported. Internal APIs use application JSON; FHIR defines the EHR boundary. Model-dependent components reach models only through the replaceable LLM adapter. Tags mark where requirements R0–R9 (Table 2) are checked; grey components were not evaluated.

string, retaining both the original text and answer-source marker. This avoids the validator's incompatible open-choice string/coded-condition exception without inventing a clinical selection. The original minimal reproduction is retained. When an answered runtime branch would lose its answer in FHIR, write-back fails closed with a non-retryable unprocessable-content response; unanswered branches do not create answers or block writing.

2.3 Synthetic inputs and model disclosure

Synthea generated the adult source cohort (Walonoski et al. 2018; The MITRE Corporation 2026). Conversion assigns fictional Taiwanese identifiers and addresses and applies TW Core profiles (Ministry of Health and Welfare, Taiwan 2025). Seeds, mappings, dropped elements, and content hashes are recorded. Taiwan-styled presentation does not establish local disease representativeness. Input validation is separate from system-output validation. The 30 author-drafted scenarios cover chest pain, headache, and abdominal pain, including expected safety stops and known rule gaps. The planned benchmark was unavailable. Separate source-level Synthea modules provide scenario-consistent backgrounds; generated patient records are not manually rewritten to force agreement.

Mechanical audits check conditions, medications, allergies, and smoking history. Human approval requires two independent physician decisions against current script hashes, including patient bindings and expected paths. At generation, 0 scripts have those approvals. Model and sampling settings and pending human QA are disclosed by role in Online Resource 1, Section S5; full prompt texts are in Sections S1 and S5. Commercial models use the Gemini Developer API; local Gemma and Qwen use Ollama (Ollama 2026; Gemma Team et al. 2025; Qwen Team 2025). Gemini API access follows the provider interface (Google 2026). Requested temperature is distinguished from the transmitted value: the Gemini adapter omits temperature for the configured newer Gemini models. Provider-effective sampling and unrecorded historical thinking settings are not inferred from defaults. Implementation, scenario drafting, orchestration, reference checking, and manuscript revision used AI assistance; the declarations specify the remaining author review.

2.4 Requirements and acceptance criteria

Table 2 maps framework requirements to checks and generated verdicts. E1–E5 denote verification activities. Only the E1 zero-error target, E3 zero code-change target, and E4 request-error stopping threshold were specified before execution. The all-match/all-pass criteria for R2–R6 and the behavioral agreement component of R7 are requirement-defined and were applied post hoc on 2026-10-14, after inspecting the results. They are not prospective clinical acceptance thresholds. R0 comes from the project principal investigator's requirement for fixed questionnaire selection; its criteria were likewise applied post hoc on 2026-10-14. Missing evidence yields an indeterminate verdict.

Table 2 Requirements, criterion sources, and generated verdicts. R0 uses E2 with E3 descriptive checks; R1/R2 use E1; R3–R6 use E2; R7 uses E3; R8 uses E4; R9 concerns E5. R7 is reported separately for each replacement in Table 5.

ID	Requirement	Criterion source	Verdict
R0	Questionnaire rules select the next question	Principal investigator requirement; criteria applied post hoc	Pass
R1	TW Core structural/profile checks	Prespecified zero errors; no validator exceptions	Pass
R2	Write/read-back identity	Requirement-defined, applied post hoc	Pass
R3	Top-level path matches the FHIR Questionnaire	Requirement-defined, applied post hoc	Fail
R4	Extracted-fact traceability	Requirement-defined, applied post hoc	Pass
R5	Scripted safety-rule execution	Requirement-defined, applied post hoc	Fail
R6	End-to-end completion without handoff	Requirement-defined, applied post hoc	Pass
R7	Configuration-only substitution retaining default behaviour	Prespecified zero code lines; default-equivalence applied post hoc	Alternate: Pass Local: Fail
R8	National peak-load capacity	No national-demand criterion available	Not verifiable
R9	MOHW hub interoperability	Access prerequisite unavailable	Blocked

2.5 Verification activities

R0 requires all saved system-model call tasks to concern chief extraction (at most once per session), report generation, or RAG translation, and each script's repeated runtime paths to be identical. Static source inspection identifies the pure next-question selector; E3 cross-model paths are reported descriptively. Usage records are bounded and explicitly incomplete, so this audit does not claim a complete history of all model calls (Online Resource 1, Sections S1 and S6).

E1 uses the pinned HL7 validator with TW Core and generated questionnaire and extension definitions (HL7 International 2026; Ministry of Health and Welfare, Taiwan 2025). R1 requires zero errors and zero validator exceptions in written resources from the templated and E2 cohorts. Structural/profile checks use `-tx n/a`; they do not establish terminology resolution. R2 requires substantive write/read-back identity after excluding specified server metadata and assigned identifiers. Missing outcomes cannot count as passes. Input validation is separate, and the supplemental E1 cohort is counted only once. Online Resource 1, Sections S3 and S7, documents these procedures.

E2 exercises invitation, text dialogue, report generation, simulated clinician confirmation, write-back, and read-back. A distinct local Qwen model plays the patient from scenario facts, following prior use of LLM-simulated patients in conversational evaluation (Tu et al. 2025; Johri et al. 2025; Schmidgall et al. 2024). Invalid constrained answers receive an explicit unknown fallback, never an expected-answer substitution. R3 compares the presented top-level runtime path with the FHIR `enableWhen` path derived primarily from the written QuestionnaireResponse, including EHR-prefilled answers. The recorded prefilled field list retains the skip context, including no-data fields omitted from QR. Nested `<linkId>.other` answer details are not separate

questions. Reconstruction from runtime answer strings and the original script-derived check are secondary evidence. Expected chief stops are excluded; unexpected stops fail. R4 requires all persisted chief-complaint facts, including those with missing evidence, to be traceable to patient utterances. R5 requires every scripted red-flag rule expectation and every non-red-flag expectation to hold. This tests rule execution, not clinical sensitivity or specificity. R6 requires write-back and an explicit `completed` or `summary_ready` workflow; expected chief stops and successful retries qualify, while manual handoff does not.

E3 substitutes model configuration while retaining executable fingerprints. R7 requires zero code changes and equality with the default configuration in R1–R6 verdicts and per-script completion, path, and red-flag results. Shared R3 representation differences and R5 rule gaps are therefore retained in the comparison baseline. This relative criterion does not require either configuration to pass every absolute requirement. One-time adapter implementation is documented separately against base commit `1b0eb1c` in Online Resource 1, Section S6.

E4 replays recorded turns with fixed-latency mock responses or real models on backend v0.18.0 with one worker. Its prespecified stopping rule is request error rate above 5%; session failure is a distinct endpoint. This local stopping rule does not define national capacity. E5 requires access to the MOHW exchange hub. Source hashes and frozen run selection generate all result macros, tables, and figures; repetitions are not independent clinical observations.

3 Results

3.1 R0: deterministic question selection

R0 is **Pass**. All 1,381 saved calls satisfy the task restriction, with at most 1 chief-extraction call per session; 150 usage records declare incomplete coverage. All 30 scripts have identical paths across 5 E2 repetitions. E3 alternate Gemini matches default paths in 30/30 scripts; Gemma matches in 26/30, with chief-stage handoffs or missed stops accounting for AP-03, AP-04, AP-07, HA-08 (Online Resource 1, Section S6). Routing sources are 100 keyword selections, 0 recorded extraction fallbacks, and 50 unrecorded sources after expected chief stops; missing sources are not counted as keywords.

3.2 R1 and R2: written resources and read-back

R1 is **Pass** and R2 is **Pass**. Across 200 templated and 150 E2 sessions, 1,347 written resources have 0 structural/profile errors and 0 validator exceptions under `-tx n/a`; all captured submissions match their read-back counterparts (Tables 3 and 4). All 350 QuestionnaireResponses have explicit outcomes, with 0 errors and 0 unverified responses. The 1,750 retained warnings comprise 1,400 Provenance terminology warnings and 350 Composition narrative warnings. The unavailable pinned participant-type CodeSystem version remains a terminology limitation, not a resolved binding. Input-cohort checks and the templated manual-handoff case are documented separately in Online Resource 1, Sections S3, S6, and S8.

Table 3 E1 written-resource verification. Input checks are excluded; exceptions would prevent acceptance.

Cohort	Resource	n	With errors	Errors	Warnings
Templated cohort	Composition	200	0	0	200
Templated cohort	Observation	197	0	0	0
Templated cohort	Provenance	200	0	0	800
Templated cohort	QuestionnaireResponse	200	0	0	0
Scenario cohort	Composition	150	0	0	150
Scenario cohort	Observation	100	0	0	0
Scenario cohort	Provenance	150	0	0	600
Scenario cohort	QuestionnaireResponse	150	0	0	0

Table 4 E1 workflow and semantic round trip. Read uses all sessions; write uses attempted writes; read-back uses returned references; round trip uses all captured submissions, including those without read-back.

Cohort	Sessions	Read	Write	Read-back	Round-trip
Templated cohort	200	100.0% (200/200)	100.0% (200/200)	100.0% (797/797)	100.0% (797/797)
Scenario cohort	150	100.0% (150/150)	100.0% (150/150)	100.0% (550/550)	100.0% (550/550)

3.3 R3–R6: questionnaire, evidence, rules, and completion

R3 is **Fail**: written-FHIR paths match in 45/100 comparable sessions, versus 100/100 using runtime answer strings. Failures occur in AP-03, AP-05, AP-10, CP-01, CP-02, CP-03, CP-04, HA-02, HA-03, HA-04, HA-06; all first divergences expect `chronic_detail` but observe `past_meds`. Patient-entered branch controls agree in 45/45, whereas EHR-prefilled controls agree in 0/55 (Online Resource 1, Section S6.3). R4 is **Pass**, with fact traceability of 100.0% (454/454). R5 is **Fail**: the failing scripts are AP-10, CP-09, exposing existing rule gaps rather than a transport failure. The full red-flag and non-red-flag denominators appear in Online Resource 1, Section S6. R6 is **Pass**, with completion of 100.0% (150/150), including 50 expected chief stops and 1 recovery after a timeout and retry. The simulated-patient fixture has answer biases, but verbatim chief complaints, separate runtime/FHIR path checks, and content-independent structure/read-back checks distinguish fixture bias from the representation discrepancy (Online Resource 1, Section S6.3).

3.4 R7: configuration-only replacement

R7 is **Pass** for alternate Gemini and **Fail** for local Gemma (Table 5). The configurations retained identical executable fingerprints with no code changes during replacement. Alternate Gemini retains the default requirement verdicts and per-script outcomes, including the shared R3 and R5 failures; this is behavioral agreement with the default, not absolute safety acceptance. The local replacement differs on AP-03,

Table 5 E3 requirement verdicts by configuration. R7 compares each replacement with the default, including per-script completion, path, and red-flag outcomes; shared rule gaps do not themselves fail this relative check. Code lines exclude one-time adapter implementation.

Check	Default Gemini	Alternate Gemini	Local Gemma 3
R1	Pass	Pass	Pass
R2	Pass	Pass	Pass
R3	Fail	Fail	Fail
R4	Pass	Pass	Pass
R5	Fail	Fail	Fail
R6	Pass	Pass	Fail
Code lines	0	0	0
R7	Reference	Pass	Fail

AP-04, AP-07, HA-08: AP-03, AP-04 enter manual handoff after chief-assessment timeouts, while AP-07, HA-08 miss expected red-flag stops. Late worker completion does not reverse a handoff. Full per-session evidence is in Online Resource 1, Sections S6 and S7.

3.5 R8 and R9: unavailable acceptance evidence

R8 is **Not verifiable** because national peak demand remains `TODO_PEAK_DEMAND`. The local request-error stop condition cannot substitute for a national-demand criterion. R9 is **Blocked** because exchange-hub access is unavailable; E5 has no measured result. Timing, turns, tokens, bounded cost estimates, fact counts, and local load observations are descriptive background in Online Resource 1, Section S6.

4 Discussion

R0 and R3 distinguish stable selection from agreement between representations. Runtime JSON uses fixed substring conditions on free-text EHR values; FHIR write-back encodes unrecognized text as coded "other" with nested original text, enabling `chronic_detail` in the published Questionnaire. These fixed rules disagree without any AI decision, even though repeated runtime paths are identical. Clinicians must decide whether out-of-option EHR text should trigger follow-up before the rules are unified; this work changes neither rule set. Passing structural and read-back checks does not establish branch equivalence or clinical meaning.

R5 failure identifies unmet rule expectations in AP-10, CP-09, including suspected pulmonary embolism and abdominal aortic aneurysm without syncope. Only chief-complaint screening runs in the default engine; later questionnaire answers do not trigger renewed screening. Other known code-level limitations include sustained exertional pressure without diaphoresis or syncope, prefill-dependent structured ectopic-pregnancy matching, and overly broad raw negation windows. These require qualified clinical review; no rules were changed.

R7 distinguishes configuration compatibility from preserved behavior. Agreement with a default that fails R3 and R5 cannot establish representation equivalence or clinical safety. Gemma's handoffs and missed scripted stops fail the relative replacement criterion. Traceability over surviving extracted facts also cannot establish extraction coverage; the retention and timeout diagnostics remain in Online Resource 1, Section S6. R8 requires externally supplied demand and a justified acceptance criterion; local load observations do not establish national readiness. R9 requires an authorized exchange-hub environment before verification can proceed.

4.1 Limitations and clinical review needs

All scenario-based verdicts are provisional pending physician review and independent transcript/fact audits. Requirements were organized after viewing the results, and requirement-defined criteria are explicitly labeled post hoc. Synthetic demographics are not representative, evidence spans do not validate clinical interpretation, and clinician confirmation was simulated. EHR prefill can retain social/contextual Condition text and English labels in Chinese forms; faithful encoding does not make that mapping clinically appropriate (Online Resource 1, Section S3). The sandbox does not establish production authorization, tenant isolation, browser SMART integration, diagnostic accuracy, clinical utility, or deployment readiness. Terminology review, institutional governance, and human approval remain separate prerequisites.

5 Conclusion

The absolute requirements R0, R1, R2, R4, R6 pass; R3, R5 fail within the defined software-verification scope. Deterministic question selection does not guarantee agreement between runtime and written FHIR branching. R7 is Pass for alternate Gemini, retaining default outcomes including shared R3/R5 failures, and Fail for Gemma. National peak capacity is Not verifiable and exchange integration is Blocked. Checks retain `-tx n/a`, without terminology resolution; simulated clinician confirmation and provisional scenarios do not establish clinical validation or deployment readiness.

Supplementary information. Online Resource 1 (`ESM_1.pdf`) provides system and prompt details (Section S1), FHIR examples (S2), synthetic conversion (S3), provisional scenarios (S4), model disclosure (S5), complete results and a provisional walkthrough (S6), harness procedures (S7), validator limitations (S8), and version locks (S9). Online Resource 2 (`ESM_2.zip`) contains the corresponding machine-readable scripts, review materials, FHIR examples, synthetic sample bundles, prompts, configurations, summaries, and a hash manifest. No scenario is physician-approved; the walkthrough remains provisional and the clinician screenshot is pending.

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Declarations

Funding, competing interests, and contributions. Authors must supply and verify these statements and final approval.

Ethics and consent. Only synthetic data were used. Authors must confirm the applicable institutional determination; no ethics approval is asserted.

Data availability. Online Resource 2 contains provisional synthetic scripts, review materials, sample TW Core bundles, FHIR examples, selected result summaries and manifests, and configuration/prompt provenance. Full session records and RAG source text are excluded. Medscape (Medscape 2026) redistribution rights require author confirmation (`TODO_AUTHOR`).

Code availability. Reproduction commands and source paths accompany Online Resource 2; the public repository location, archival identifier, and code/data licenses remain `TODO_AUTHOR`.

AI assistance. Claude Code and Codex assisted code and scenario drafting, verification orchestration, source checking, and manuscript drafting and revision. The authoring log records Claude Code with `claude-opus-5-5` and Codex in the GPT-6 family; exact tool releases and Codex runtime version remain unrecorded. Authors must confirm these details and the final disclosure wording (`TODO_AUTHOR`; Online Resource 1, Section S5). Authors must verify citations, review technical and clinical content, and approve the final text; scenario approvals and independent audits remain pending.

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